LOAN APPROVAL PREDICTION

Data Science Internship Project Report

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Domain: Data Science
Course: Third Year Engineering – Information Technology (TE IT)

Project objective: analyze applicant data and train a machine-learning classifier to predict loan approval status.

1. Introduction

Loan approval prediction is a classification problem that uses applicant information to estimate whether a loan application is approved. This project explores a loan eligibility dataset and trains a Random Forest classifier for educational purposes.

2. Objectives

Inspect and clean applicant data, analyze patterns through visualizations, train a classification model, and evaluate it using accuracy, precision, recall, F1-score, and a confusion matrix.

3. Dataset Description

The dataset contains 614 records and 13 columns. The target variable is **Loan_Status**: Y means approved and N means not approved. Input columns include applicant income, co-applicant income, education, employment status, loan amount, credit history, and property area.

4. Tools and Technologies

Python, Pandas, Matplotlib, Seaborn, Scikit-learn, and JupyterLab.

5. Data Preprocessing

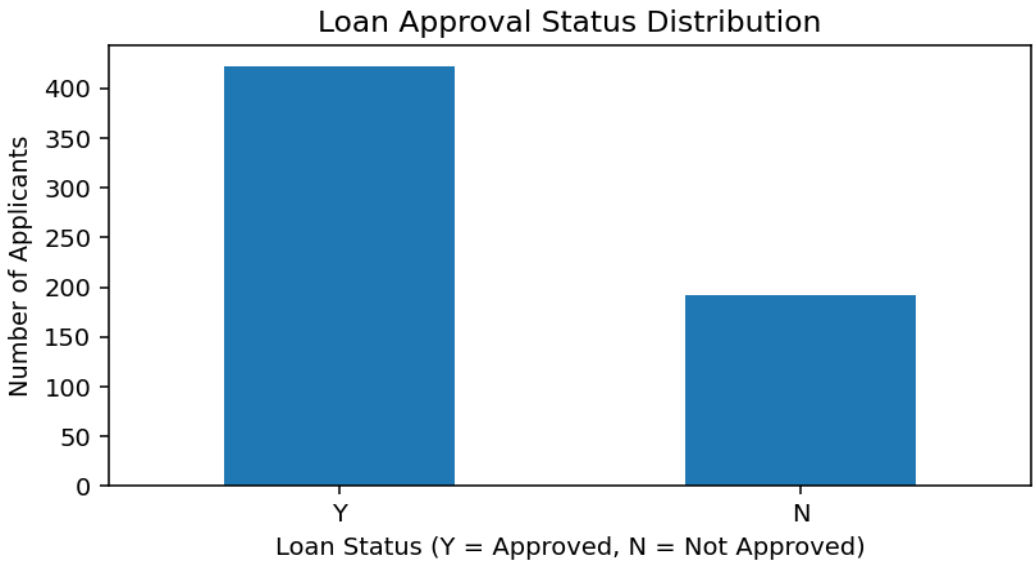
The workflow removes duplicate rows, fills missing categorical values with the mode and missing numerical values with the median, removes Loan_ID from model inputs, and converts categorical predictors using one-hot encoding. Loan_Status is encoded as 1 for Y and 0 for N. The data is split into 80% training and 20% testing with stratification.

Note: For production work, fit preprocessing only on training data using a Pipeline to help prevent data leakage.

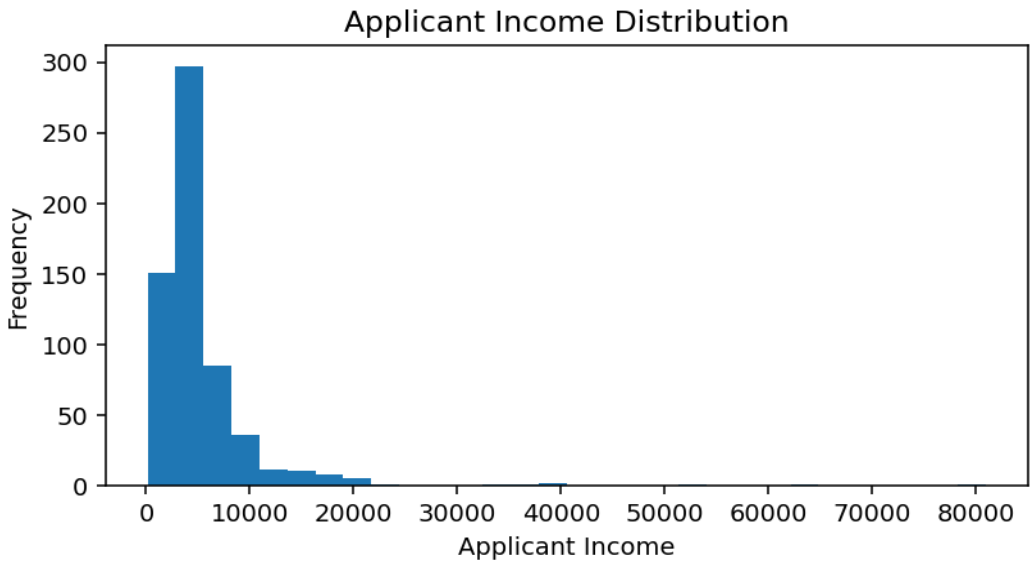
6. Exploratory Data Analysis

Charts generated from the supplied dataset:

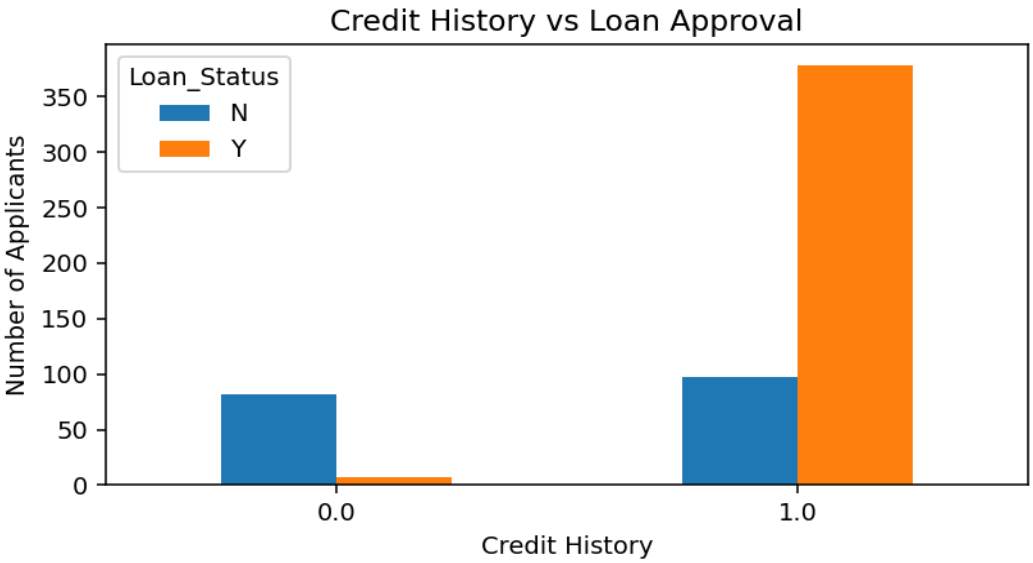
Loan Approval Status Distribution



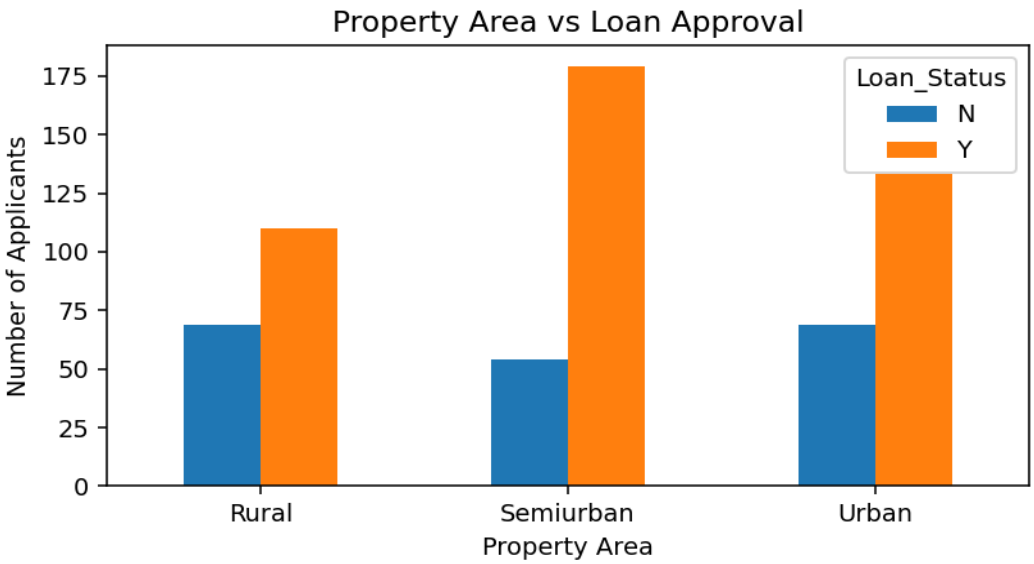
Applicant Income Distribution



Credit History vs Loan Approval



Property Area vs Loan Approval


Property Area vs Loan Approval
Loan_Status
N
Y
Number of Applicants
175
150
125
100
75
50
25
0
Rural
Semiurban
Urban
Property Area

7. Machine Learning Model

A Random Forest Classifier with 100 trees was trained using the preprocessed training data and evaluated on a held-out test set.

8. Model Results

Metric	Result
Test set size	123 records
Accuracy	83.74%
Correct predictions	103
Incorrect predictions	20
Approved-loan precision	0.85
Approved-loan recall	0.93
Not-approved-loan precision	0.80
Not-approved-loan recall	0.63

Confusion matrix (actual rows × predicted columns):

	Predicted: Not Approved	Predicted: Approved
Actual: Not Approved	24	14
Actual: Approved	6	79

Interpretation: 103 of 123 test cases were classified correctly. Recall for approved loans was 0.93, compared with 0.63 for not-approved loans.

9. Conclusion

The Random Forest model achieved 83.74% accuracy on the test set. It performed better at identifying approved loans than applications that were not approved. Further work could compare algorithms, use cross-validation, tune hyperparameters, and evaluate fairness.

10. Limitations and Responsible Use

This dataset is relatively small, and results depend on the data split and preprocessing choices. Loan decisions are high-impact financial decisions; this educational model should not be used as the sole basis for real lending decisions.

11. Project Files

Recommended repository contents: Loan_Approval_Prediction.ipynb, README.md, visualization PNG files, and this report PDF. Include the dataset only if permitted and within repository size limits.

12. Acknowledgement

I thank Alfido Tech for the opportunity to work on this Data Science internship project.